

Outline for Research Project on Sales Forecasting

Topic: Improving Sales Forecasting with Machine Learning

Research Idea: This research explores how machine learning can make sales forecasts more accurate than traditional methods. Traditional forecasting often misses important patterns in sales data, leading to issues like overstocking or running out of products. By using advanced machine learning techniques, this study aims to develop better forecasting models that handle complex data patterns more effectively. I picked this topic because it can help better forecasts and can help businesses manage inventory and resources more efficiently.

Problem Statement: Sales forecasting is crucial for managing inventory, but traditional methods are often inaccurate and fail to capture complex data patterns. This research will use machine learning to improve forecast accuracy, helping businesses avoid problems like excess inventory and stockouts. Better forecasts can lead to cost savings and better decision-making for businesses.

Data Exploration: The study will use many sales data, market trends, and economic indicators. Data will be collected from company records and public sources. Machine learning models will be applied to this data to create more accurate sales forecasts.

Knowledge Transfer: The research will produce improved forecasting models that businesses can use. It will provide practical advice on applying machine learning to sales forecasting, helping businesses make better decisions and manage their inventory more effectively.